

THE SILENT WATCHERS:

The Critical Importance Of Securing Satellite Communications

Satcom powers everything from high-speed internet to disaster response—yet its growing exposure to cyber threats raises urgent questions about security and resilience.



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In the silent vastness of space, satellites drift unnoticed above Earth, but their role is indispensable to billions. These technological marvels, orbiting thousands of kilometres overhead, act as the unseen sentinels of civilisation. They beam internet to remote villages, track hurricanes before landfall, and guide aircraft safely across oceans. But behind this façade of seamless service lies a growing threat—one that could dismantle the very foundations of global connectivity and security if left unaddressed.

Dark skies hide more than just stars—they mask an invisible war fought every second to protect the signals powering

modern life. A stark reminder came in 2019, when GPS disruptions swept across Scandinavia, leaving pilots disoriented, ships adrift, and smartphones misaligned. Initially traced to suspected Russian interference, the incident revealed a sobering truth: satellite communication systems are far more exposed than many realise. What was once a technological luxury has become a critical infrastructure. Satellite communication now underpins everything from global internet and defence networks to emerging technologies like 5G non-terrestrial networks (NTN). Yet, as these systems expand, so does the risk of increasingly complex and targeted attacks.

Fortress in space: Securing satellites from within

Security at the system level within satellites themselves remains critical. This entails implementing robust physical and cyber security measures to protect onboard computers, communication equipment, and sensitive data. These include:

Hardware security. Employing tamper-proof hardware, secure boot mechanisms, and hardware-based security modules

Software security. Enforcing secure coding practices, conducting regular security audits, and patching software vulnerabilities promptly

Physical security. Protecting satellites from unauthorised access and tampering during manufacturing, launch, and on-orbit operations

